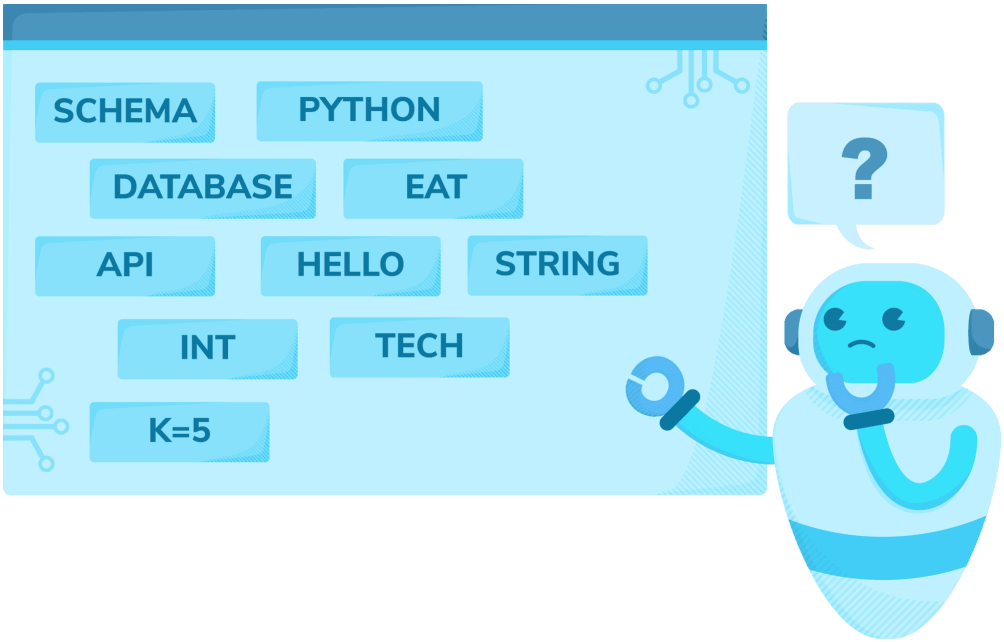
Generation

Explore the OpenAI Completions endpoint to generate text and code using Python. Understand how to craft prompts, adjust temperature for creativity, and use penalties for diverse outputs. Learn to apply these capabilities in real-world scenarios such as AI chatbots and content generation.

Overview

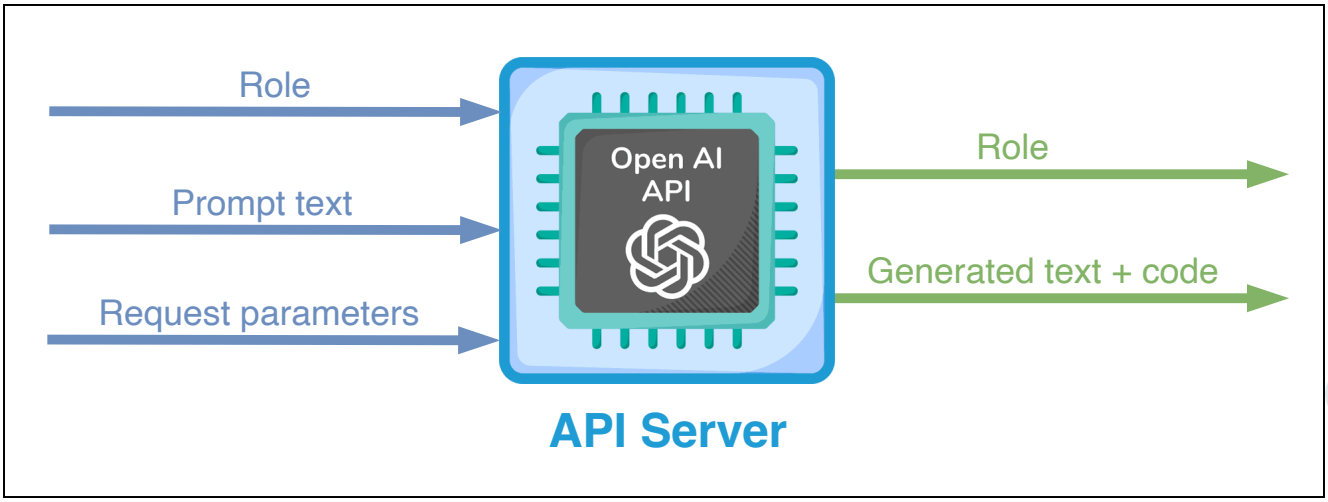
The OpenAI models are also great at generating code and text as they have been trained on a large amount of data from different resources.



Ask

Generation

We input the prompt text and request parameters into the OpenAI API's model to generate both text and code.



Completions: text generation

Generate text

Now that we have learned about how OpenAI API's model works. We can utilize the following prompt to generate new ideas about deep learning and images:

```
Brainstorm some ideas combining deep learning and images:
```

Example

Let's try the above example in the code widget below:

Please provide values for the following:

SECRET_KEY Not Specified...

Edit Import Values from JSON

Ask

main.py Python

```
# Generation

from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

completion = client.chat.completions.create(
  model="gpt-3.5-turbo",
  messages=[
    {"role": "system", "content": "You are a helpful assistant."},
    {"role": "user", "content": "Brainstorm some ideas combining deep lea
  ],
  temperature=0.75,
```

Run

Generating creative text

In the code above, we use `temperature` with a value of `0.75` to include 75% randomness for text generation, `max_tokens` as `200`, which is the output text's maximum length, and `top_p` with a value of `1.0` to include all the tokens in new text generation. The `frequency_penalty` and `presence_penalty` are two ways to ensure that the new text has diverse words, and both are used with the values of `1`. The `frequency_penalty` decreases the chance of repeating words, while the `presence_penalty` increases the chance of including new words.

Try out all the engines mentioned in this lesson and examine the different effects on the output text.

> **Note:** We can increase the quality of the results, if necessary, by giving some examples in the prompt.

Let's take another example where we ask the API to list the current trending business ideas for IT. The prompt for this can be:

Ask

```
List the current trending business ideas for IT.
```

Now, ask the API to list the future business ideas for IT. The prompt for this can be:

```
List future business ideas for IT.
```

Use a greater value of `temperature` to produce creative ideas.

Generate code

We can generate code in any programming language using the chat completions endpoint. For instance, we can generate a C++ program to print `"Hello World"` using the following prompt:

```
Write a Hello World program in C++.
```

Example

Let's try the above example in the code widget below:

Please provide values for the following:

SECRET_KEY Not Specified...

Edit Import Values from JSON

main.py Python

```
# Generation

from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

completion = client.chat.completions.create(
```

Ask

```
    model="gpt-3.5-turbo",
    messages=[
      {"role": "system", "content": "You are a helpful assistant."},
      {"role": "user", "content": "Write a Hello World program in C++."}
    ],
    temperature=0.75,
```

Run

Generating code

In the above code, we use `temperature` with a value of `0.2` to make the output more focused and deterministic.

> **Note:** Try to change the programming language in the prompt and check the output.

Practical use case

A practical use case for OpenAI's text generation capabilities is an AI-powered chatbot or help assistant. Imagine a customer support chatbot integrated into a company's website or application. This chatbot is designed to assist users with common queries, provide troubleshooting guidance, and offer product recommendations. By leveraging OpenAI's text generation models, the chatbot can engage in natural and contextually relevant conversations with users, enhancing the overall user experience.

Mark As Completed

Ask